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PI Name:	Dunn-Rankin, Derek Ph.D.		
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PI Email:	ddunnran@uci.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	949-824-8745
Organization Name:	University of California - Irvine		
PI Address 1:	Department of Mechanical & Aerospace Engineering		
PI Address 2:	4200 Engineering Gateway Bldg, EG3224		
PI Web Page:			
City:	Irvine	**State:**	CA
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Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
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	NOTE this is a successor agreement to "Applications of Electric Fields in Microgravity Combustion Science (NNX07AB55A)," per D. Stocker, NASA Glenn Research Center. This joint project between the University of California, Irvine and Imperial College, London explores the use of large electric fields in combustion for the purpose of improving the performance of energy conversion systems. As an actuator, electric fields can affect flame stability and soot formation; while as a sensing device, the electrical response at saturation is a flame characteristic. In addition to preparing for the proposed ACME E-FIELD experiments aboard the International Space Station (ISS), we are currently studying electric field effects on quenched jet flames, partially-admixed air/fuel jet flames, and their associated dangers. This report summarizes recent findings. Specific activities include:		

Task Description:	1.) Study electric field control of flames with wall effects near quenching. The experiment involves flow field imaging, along with OH chemiluminescence and OH planar laser induced fluorescence (PLIF) techniques. Measurements include thermal emission and carbon monoxide emission, and analyses evaluate thermal emission. 2.) Characterize time-scales related to the interaction between ions and neutral molecules within the burner/mesh electrode gap. These time-scales provide information into how an electric field affects combustion related parameters. These times-scales were determined using high-speed schlieren imaging and ion current/applied voltage measurements. Our understanding developed in these areas may improve kinetic mechanisms for diffusion flame modeling (including charged species), help clearly identify the risk of CO emission from quenched flames, and help determine if electric fields can actively control the release of carbon monoxide from flames. Task Description: The overall goal of this project is to understand the behavior of chemi-ionization in flames and to demonstrate the use of this understanding to manipulate combustion in beneficial ways (including enhancing flame stability and reducing flame soot). The project uses detailed measurements of electrical properties of flames in several ways. First, we can create a neutral wind (ion driven convection) locally at the flame front; this is one of the rare and important action-at-a-distance effects on flames that can be used as a localized actuator. The effect of the wind has been demonstrated with a candle flame in parabolic flight-based microgravity and in a spherical cage drop tower, but there has been no systematic study of the ion wind effect in the absence of the confounding effects of the approximately equal influence of buoyancy. Second, we can use electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions. It is possible, therefore, to modify soot formation and to use charges attached to soot to affect soot transport. By adjusting the soot trajectory through the reaction zone it is possible to affect the oxidation of soot to prevent its eventual release or to modify the residence time of soot in different zones of the flame. It is also possible to use the neutral ion wind to change the mixing behavior of the flame and reduce soot in this way. Finally, we can use the electrical properties to characterize the flame – there is a quantifiable relationship between the carbon content of a fuel and the ion current it can produce at saturation. This relationship is already exploited in a typical flame ionization detector (FID) analyzer for hydrocarbon emission where the carbon is burned in a hydrogen flame to create an ion current in proportion to the carbon influx. In direct flame interrogation, however, there are several confounding variables that can affect the ion current, including flame temperature (which generally affects the O atom concentration) and the reaction zone volume. Flame temperature can in turn be affected by levels of premixing, diffusion rates, fuel composition, and radiative loss. It is possible, therefore, to determine flame character from the ionization curve response. Task Significance: Understanding electric field interactions with flames is central to combustion control, particularly in situations near limit operation of the flame where small effects are amplified dramatically. Electric fields have been shown to modify burning rates as well as to enhance and reduce flame propagation even to the point of flame extinguishment. Our experiments provide a unique ability to make simultaneous measurements of both physical and electrical properties of flames.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The control of combustion has the potential to improve efficiency and reduce emissions from burning fuels. Since high power density often requires combustion, these improvements will be important no matter what the fuel source. Electric fields acting on flames have the potential to aid in combustion control. In addition, electrical properties of flames can identify poor performing boiler flames that release poisonous carbon monoxide. Our studies show that a flame's electric signature can capture incipient quenching before dangerous emissions result. Understanding the links between electrical character and flame behavior may allow improved sensing of poor performing combustion systems.
	In addition to preparing for the proposed ACME E-FIELD experiments aboard the ISS, we are currently evaluating an experimental method of measuring ion number density in a flame using a Langmuir probe, the dynamic interaction between the flame and chemi-ions, and the use of electric fields to control flames impinging on a surface. This report summarizes recent findings. Specific activities include: 1.) Study the electrical behavior of a non-premixed co-flow flame. This is done by measuring the ion current that is collected by a Langmuir probe placed into the flame. The ion current gives insight into the production and transportation of chemi-ions due to a natural occurring methane combustion process without the effects of an external electric field. 2.) Study the dynamic interaction between the flame and the electrical body force created by a flux of chem-ions. This is done by comparing transient flame behavior using a high speed camera and the transient ion current collected when an electric field is turned on. 3.) Study electric field control of flames with wall effects near quenching. The experiment supplements our earlier measurements of OH PLIF with 2-line OH temperature measurements and 2-photon CO PLIF to confirm the findings that electric fields can moderate carbon monoxide release from flames impinging on surfaces. Summaries of these findings appear in this report with the details of all the work appearing in the publications and conference proceedings identified in the Bibliography section. This year the project produced 1 peer-reviewed journal article (with 2 others in preparation); 12 conference papers/presentations; 4 M.S. thesis documents; and 1 Ph.D. dissertation. All ground-based experiments were performed using a coflow burner that is dimensionally similar to the proposed ISS coflow burner and that uses a fuel jet matching the dimension of the ISS jet burner. The burner consists of two stainless steel plenums threaded together. Fuel, introduced in the bottom plenum, is passed through a stainless steel tube (2.13 mm ID) and exits the top surface fully developed. Air entering the top chamber passes through flow straightening beads

and a honeycomb mesh. The air exits the top surface with a top hat flow profile at the exit. A variable high-tension power supply connected between the burner and downstream mesh electrode produces the applied electric field. The electric field distribution is deformed by the flame's space charge. Regardless, defining an applied field strength, E = V/D, is a useful approximation when describing experimental results. Ion current is computed using Ohm's law and the measured potential drop across a shunt resistor placed between the electrically isolated burner and the building ground. An acrylic chamber shields the flame from room currents. A Faraday cage was also added to isolate the measurements from EMI radiating outside of the experimental volume.

(1) Langmuir Probe Studies

The object of this study is to evaluate the method of using a Langmuir probe to determine the ion number density produced by the flame. The short-term goal is to determine the naturally occurring ion number density (without an external field) while the long-term goal is to apply the Langmuir probe study to a flame with an external field. With the ion number density in the presence of an external field known, the electrical body force (ion-wind force) can be calculated.

Electric probes have a long history within the field of plasma diagnostics. They are simple to build and are easily implemented into plasma experiments. However, the results they provide are not so easy to analyze. A particular type of electric probe, the Langmuir probe, has found its way into combustion research in order to study naturally produced flame ions. In the present work, a small potential is applied to a Langmuir probe with respect to ground (burner). This procedure is known as the single-probe method.

For the geometry of the plasma (i.e., the flame) in this experiment, the best choice is to use a spherical tip Langmuir probe because it will have the minimum impact on the heated flow of gas inside the flame and because it provides a relatively simple means of calculating the ion flux from the measured ion current. The spherical shape reduces the complexity of the problem studied.

In combustion environments, CH* is predominately formed from C2H and atomic oxygen. Previous E-FIELD Flame investigations have suggested that there exists a high correlation between CH* and H3O+. However, this was shown at a global level. The current results show the correlation (if any) between CH* and ion current collection locally.

Probe System

A co-flow burner was used for creating a small methane/air diffusion flame. A simple Langmuir probe was used for ion current detection. It was made of stainless steel and insulated with a tube of alumina silicate to avoid melting, electric arcing, and electric transmission. Stainless steel was chosen because of its low resistivity and high melting point (1680 K).

The probe dimensions are a length of 6 inches (152.4 mm) and a tip of 1 mm of diameter. The probe was attached to two motor driven Velmex sliders for 2-axis control. One axis refers to the vertical axis Z (symmetry axis of revolution) and the other to the radial position r. The probe was swept radially from R=-19 mm to R-19 mm for different heights. However, an ion current was only detected within the range of R=-6 mm to R=6 mm, representing the extent to which the charged species can diffuse from the flame sheet radially. Positive potentials of 40 V, 80 V, and 120 V (generated by the high performance laboratory grade device: TREK Model 609A-3) were applied to the probe to attract negative charge carriers and to repel positively charged species.

Probe potentials that are too high run the risk of creating ion winds, which would disturb the flame from its natural state. It was found that at potentials above 400 V, the flame began to bend. The generated ion current that flows through the shunt resistor to the ground was measured and related to voltage using Ohm's Law. From preliminary tests, it was determined that a Faraday cage was necessary to isolate the probe from external noise. The flow rate used was 26.9 mL/min. The ion current signal shows good symmetry across the flame with the peak in the reaction zone.

Chemilumenscence System

Chemiluminescence is radiation emitted from electronically excited molecules as they return to their lower energy state. The wavelength of the radiation that is emitted is characteristic of the particular molecule and the particular transition the molecule undergoes. Since the amount of radiation observed in the flame at a particular wavelength is proportional to the concentration of the associated excited molecule, a measurement of the radiation can be directly related to the concentration of the excited molecules.

In hydrocarbon combustion, the most common electronically excited species are OH*, CH*, and C2*. In the early 1970s chemiluminescence was identified as a marker for heat release, reaction zone, and equivalence ratio, thereby providing a relatively easy diagnostic alternative for the measurements in practical combustion applications and an inexpensive diagnostic tool that avoids expensive instrumentation, e.g., lasers. The present work is focused on the detection of CH* at a wavelength of 431 nm.

A bandpass interference filter was used to collect the appropriate wavelength range of the light emitted from the flame. The present work is focused on the emission of light from CH* which is a known precursor for positively charged species such as CHO+ and H3O+. H3O+ is known to be the only positively charged species that is able to survive outside the flame to any substantial extent. Since CH* is an accepted precursor to ion formation it is important to describe one of the paths for the formation of H3O+ from CH and CH*. CH* is produced chemically via the reaction of the ethynyl radical with monatomic and diatomic oxygen. Destruction reactions occur by spontaneous emission and by collisional quenching. There are two reactions that give rise to chemi-ions. The second reaction is more likely to occur in flames if the concentration of the excited CH can exceed the thermal equilibrium populations by several orders of magnitudes. Finally, H3O+ is formed from CHO+ by the addition of water. Pictures were taken from the flame and then post-processed with Matlab to compute the intensity.

Results and Discussion

The objective of this experiment is to determine what height above the nozzle produces the maximum collection of ion current. The probe swept the space several times at different heights from H=0.5 mm to H=6.5 mm with increments of delta H=1 mm. The potential applied to the probe is 40 V, 80 V, and 120 V and three different flow rates are used (26.9 mL/min, 40.4 mL/min, and 53.8 mL/min).

Images were taken of the flame using a 430 nm bandpass interference filter to capture the luminosity from the flame due

Task Progress:

to CH*. Two images were also taken using 440 nm and 420 nm filters. The purpose for these two extra images was to determine if there was any contribution to the detected light from any source other than CH* (e.g., soot). The results of these images concluded that the 430 nm filter was capturing only CH* chemiluminescence. Several pictures of the flame are taken and then averaged.

A three-point inverse Abel deconvolution method is applied for deconvolving CH* chemiluminescence images. Assuming that the plasma sheath created by the Langmuir probe is much smaller than the dimension of the probe, the results show that the ions have the highest concentration in the flame (compared to the location of CH*) but that they are transported far outside the flame.

A pixel intensity map is compared with ion current peaks position and amplitudes, and ion current peak positions and are gathered and correlated with pixel peak position and pixel intensity peaks for the corresponding heights (from H=0.5 mm to H=4.5 mm) for the three sampled probe potentials. Ion current and pixel intensity values are normalized by their corresponding maximum value. From these experiments it can be seen that the overall maximum correlation is achieved for Vp=40 V and for H-1.5 mm to H=3.5 mm; above H=3.5 mm this position correlation disappears. As the probe voltage increases, the location of peak ion current moves outward away from the CH* band. This is reasonable since higher probe voltage encourages the charged species to move toward the probe. In addition, as the probe potential increases the correlation is also reduced. There exists a poor correlation between the pixel intensity and the collected ion current from the probe for different probe potentials except for H=1.5 mm where the correlation is reasonable. The probe potential which best correlated is for Vp=40 V and for H-1.5 mm. The results also suggest that the ions are evenly distributed throughout the flame region.

Calculating Ion Number Density

For a spherical probe, the conservation of charge is mathematically expressed as a gradient of ion flux. Then, a relationship between the charge flux anywhere in the plasma sheath can be related to the charge flux at the sheath edge. The velocity of ions entering the plasma sheath is approximated to be the Bohm velocity. Combining these equations, the number density of chemi-ions as a function of ion current can be expressed. The only unknown left is sheath diameter. A thin sheath approximation can be used. Using the characteristic values consistent with the experimental parameters, it is found that there is sufficient evidence that a thin sheath approximation can be used. It is found that in the region of highest ion current collection by the Langmuir probe the ion number density is close to that found by others.

While this simple analysis produces reasonable results, it is very well known that Langmuir probe theory is notoriously very complicated and an analysis for the case that the plasma sheath is not thin should be performed. By inspection of the experimental results it could be seen that the maximum possible sheath diameter that could be produced by the flame plasma/Langmuir probe interaction is 5 mm. Using this in calculation produces an ion number density of around the same order.

(2) Dynamic Flame Response

The application of an external electric field to a flame is well-known to effect the flame's behavior. Previous E-Field Flame studies have visualized the dynamic effect an electric field has on the flow field by using high-speed schlieren imaging. These studies revealed the time-scales involved during the process of manipulating the flame with an external field. The results of the time-scales suggests that the full process can be qualitatively described by the following steps:

i) Ions are immediately collected (1-2 ms after the electric field is initiated) at the electrodes. However, the full ion flux that is produced by the field is not reached until 8-10 ms after the field is initiated. The cause of this time difference is most likely that the collection of ions is being restricted by the flame's own plasma sheath adjusting to the external field. This flame sheath behavior is not a new concept, however, is rarely accounted for in electric field flame studies in the literature.

ii) The flow field was seen to begin to change approximately 2-3 ms after the field was initiated. Even though the full ion flux is not reached within this time-scale, enough ions are extracted from the flame to produce an ion-wind force sufficient enough to manipulate the bulk gas. The difference in the two time-scales indicates the time it takes to produce enough ion/neutral collisions to produce this ion-wind body force. The change in the flow field was shown to originate near the base of the flame and progress its way up downstream from the burner exit.

iii) It was also shown from high-speed imaging of the flame itself that the first signs of the flame being manipulated by the field is 2-5 ms after the field is initiated. This suggested the flame is reacting to the change in the flow field and not by electric field/chemi-ion flux directly.

Steps i-iii describe the initial steps of how the electric field is able to manipulate the flame. What was also shown from these studies was that the flame/electric field interaction is not a simple one-direction process. This is because the flame is not only the object being controlled by the electric field but also the source of ions (i.e., the source of the ion-wind). In fact, the time-scale involved with the entire environment going from its initial state to final state can be described by these three major elements (electric field induced chemi-ion flux, neutral gas flow field, and flame/chemi-ion source) reaching an equilibrium state with each other.

A final interesting result from these studies was the existence of field strengths that produced final states that were non-steady and oscillatory in nature. It was observed that at high enough fields the transient ion current measurements and the transient thermal plume boundary began to oscillate with a frequency 74hz. This unsteady behavior is exemplary of the how the three elements described above are coupled above.

This objective of the current study is to characterize the unsteady behavior at high electric field strengths to determine what are the important parameters controlling the oscillations.

Determining Flame Height from Flame Videos

Further evidence of flame dynamics was obtained from high-speed video taken before and after the electric field was initiated. Each video was recorded at 500 frames per second (fps). Higher than 500 fps, the small amount of light produced by the flame is too weak to capture videos sufficient enough to analyze. For each video, an LED light is used to time the moment that the electric field is initiated. The time indicated by t=0 ms is the defined to be the frame just before the LED light turns on (in other words, the first frame in the video that the LED is turned on is considered t=2 ms).

Because the light from the flame was weak, light filters were not used to isolate at a particular wavelength. Hence, the camera also captures light emitted by soot particles. This is not easily noticeable from the black and white images but it does add difficulty in defining the location of the flame height based purely from pixel intensity. Even when soot does not seem to be present, defining flame height based on maximum pixel intensity, in many cases, produces pixel locations significantly far from where the human eye would locate the flame height in the image. A more reliable method was to choose the maximum gradient of the pixel intensity along the centerline of the burner. A 5-point moving average smoothing curve of the pixel intensity further cleaned up the flame height location. For selected videos, using the gradient of smoothed pixel intensity gave pixel location results no more than 2-3 pixels from where the human eye would define the pixel height location. After calibration (calibration factor of 70 microns/pixel), this results in a -0.2 mm difference from the location determined by the human eye. Also, it should be noted that when the electric field is turned on, the flame luminosity increases, which makes the pixel intensity profiles of each frame appear more distinct.

Transient Ion Current Measurements

A shunt resister was wired in series in between the burner and high voltage power supply. The transient ion current was determined by measuring the voltage drop across the shunt resistor and using Ohm's Law to determine the current. The ion current was sampled by at data acquisition at 1000 samples per second. The timing of the ion current measurements was defined similar to the flame height. The time, t=0 ms, is defined as the sample reading just before the command signal is sent to the high voltage power supply (i.e., t=1 ms is the sample in which the data acquisition reads the first high voltage measurement). Because the data acquisition has a better temporal resolution than the high-video camera, the timing of the data acquisition is considered to be the baseline timing source. Assuming the LED emits light much less than 1 ms after it feels a potential difference across its leads, t=0 ms is known within 1 ms between both the high-speed videos and the ion current measurements. Based on this experimental unknown, this error (-1 ms) would shift the transient flame heights up to 1ms to the left from their current temporal locations.

Results and Discussion

The oscillations in ion current and luminosity were faintly observable in the measurements at voltage steps as low as -6.1 kV but were not observable in the flame height measurements until -6.4 kV. This is because the small random fluctuations in the flame height position (±0.12 mm from the mean flame height) exceeds the amplitude change caused by the electric field below -6.4 kV. An amplitude of about 0.05 μA (±0.05 μA from the mean ion current) causes a significant enough effect for the oscillations to be observable in the flame height. As the intensity of the voltage steps increase above -6.3 kV, the amplitude of ion current oscillations increases, as well as, the amplitude of the flame height oscillations up till -6.8 kV. This step voltage produces the largest oscillations in the both the ion current and flame height. At a step voltage of -6.9 kV, the oscillations begin to disappear.

The 74 hz oscillations are easily observable. The amplitude of the ion current is ±0.1 μA from the mean ion current and the amplitude of the flame height is ±1 mm from the mean flame height. It is also interesting to point out that when the flame height reaches a high value the magnitude of the ion current is also high but out of phase by 1ms behind the flame height. This is most likely because a higher flame height decreases the distance between the flame and the downstream electrode (i.e., lower resistance), therefore, increasing the ion current.

As stated above, the uncertainty in the temporal calibration of the flame height produces a phase shift in the flame height profile up to 1 ms to the left. Accounting for this uncertainty, the ion current is between 1 and 2 ms behind the flame height. This suggested that (assuming the transportation of ions from the flame to the downstream electrode is the time limiting mechanism) when ion flux from the flame is increased (or decreased) it takes between 1-2 ms for it to be transported to the downstream electrode in order for an ion current to be measured by the shunt resistor. The hypothesis that the electrical resistance can be treated as a function of flame height for increasing/decreasing the ion current, seems reasonable enough to explain the unsteady ion current vs flame height profiles but does not explain what causes the flame height to oscillate. In order to do this, further studies of unsteady schlieren behavior is needed.

(3) Impinging Flames

There were two major accomplishments regarding the effect of electric fields on impinging flames subsequent to the last annual report. The first accomplishment is the use of 2-line OH thermometry to non-intrusively measure the temperature profile of the flame even when under the influence of an electric field. The second accomplishment was to use 2-photon planar laser induced fluorescence (PLIF) to measure the concentration of carbon monoxide (CO) in the impinging flame. The impinging flame study was the topic of Dr. Yu-Chien Chien’s Ph.D. dissertation completed as part of this project.

As in all prior studies, the experiments are conducted with a coflow burner and a 2 mm thick stainless steel plate progressively lowered toward the burner surface so that it gradually quenches the flame. The stainless steel coflow burner is 13 cm tall and has a 4 cm outer diameter. At the exit, the burner has a 2 mm inner diameter center tube carrying fuel, and air is provided separately through a concentric outer ring. The air is designed to have a uniform laminar flow distribution at the exit after it passes through a bed of beads and a honeycomb mesh close to the exit. Inside the burner, the length of the tube is sufficient to ensure fully developed laminar flow at the exit for the range of desired Reynolds numbers. The metal plate used as the quenching surface is 50 mm square. It is indirectly connected onto a vertical-axial translation stage through two cylindrical ceramic threaded posts. The diffusion flame used for this work had flow rates equivalent to a constant nominal speed of 20 cm/s for both methane and air at the burner exit.

The 2-photon CO signal depends on a wide range of parameters. Although complex, many of these values can be determined, at least for relative conditions, permitting CO fluorescence to provide a clear qualitative view of the spatial distribution of this important molecule. The wavelength chosen for the experiment is 230.034; the wavelength was monitored and calibrated with a Xenon lamp and a 1-meter spectrometer. This wavelength belongs to the transitions in the Q branch with a rotational quantum number around J=11.

A Nd:Yag laser offering 850 mJ at 1064 nm, provided, after doubling and mixing, 250 mJ at 355 nm with 5 ns duration at FWHM. The pumping energy at 355nm goes into a dye laser (Vista) with Coumarin 460 diluted in methanol. After the laser energy at 460nm is doubled and separated from the fundamental with a Pellin-Broca prism. The system is optimized for maximum output energy at 5 mJ per pulse. The laser beam is converted into a laser sheet using a quartz 100 mm focal length convex cylindrical lens that focusses the light into the burner. The lens has been chosen as a tradeoff between the energy density of the flame and the Rayleigh range, both of them have to be maximized for a good signal. For correcting the images for the intensity distribution of the laser sheet, a gas cell with CaF2 windows and

estimated concentration of CO around 300 ppm was used as a uniform sample target. The detection system used to take the images is an emICCD PI-MAX4 with a 75 mm lens and a filter centered at 486 nm with a 50% transmission window of 10 nm. The gate for the images is 100 ns.

Three positions were studied with the burner at different distances with respect to the plate. An average of 1000 frames was taken for each position for both the laser intensity profile and the fluorescence signal. The cylindrical lens was selected to be 100 mm focal length in order to stay in the linear regime of the CO fluorescence, that is, the regime where the CO ionization dominates. The beam waist is estimated to be about 50 µm in thickness, and the average power density is approximately 0.5 GW/cm2. This power density is enough to ensure that ionization dominates the depopulation of the upper state, giving an approximate linear behavior with intensity for atmospheric pressure conditions.

The two-line OH thermometry measurements use the same apparatus but a different dye system and wavelengths. The basis for two-line OH temperature PLIF is the ratio of signal strength from two spectral lines. Where the ratio at every pixel of the camera is a function of several constants, temperature, laser intensity, both overlap integrals, and the ratio of both quantum fluorescent yields. The spectral overlap between laser profile and absorption line profile is assumed to be the same for both transitions, which is a source of a small systematic error, even though the laser profile is very similar for both transitions. The linearity of the signal was checked by comparing the overall integrated signal of the fluorescence images from the camera with the average energy power from the laser used for the images. The transitions selected from the A-X (1,0) band are the P1(3) and P1(10) around 283 nm.

200 images are averaged before ratioing to obtain a better signal to noise image. The averaged image is weighted with the laser intensity mapping obtained from a fluorescent dye diluted mixture. The images are aligned one by one since the laser intensity mapping is taken for every transition due to nonlinear behaviors in the doubling processes. The images are corrected using the average of the images (instead of 2 images) by using the overall integrated signal for each sequence of pictures. Finally, isolating temperature from the previous equation, a temperature ratio is obtained for each pixel. In order to only obtain meaningful temperatures, a product of both averaged intensity transitions is computed, and a constant is used to limit the low end in the not meaningful temperature data, assigning 0 K where it is not meaningful. The lower temperature limit is 1000 K because OH only exists at temperatures above 1000 K. All the measurements are repeated for different relative positions of the flame with respect to the plate.

Key results of the study show images for similar flame conditions of: two-photon CO, chemiluminescence, and two-line OH PLIF temperature measurements. Qualitative two-photon CO PLIF measurements in a non-premixed coflow impinging flame with a plate downstream the flow were obtained. The temperature dependence of the wavelength used has been studied in order to understand the images better. The intensity dependence of the signal has been corrected and additional corrections have been applied in order to account for the complex intensity dependence of the fluorescence signal. Contribution from C2* in the flame studied are larger than in premixed flames, but distinct images of CO without C2* interference were obtained through averaging and a narrow bandpass filter.

These results show that absolute flame temperature is not affected by the plate location but the extent of the hot zone of OH is affected. In addition, it is clear that when the plate is very close to the burner that CO is able to escape along the surface of the plate rather than being confined within the flame zone. Future work is to confirm the relationships between OH location and temperature, the boundary layer flow along the plate surface, and the release of carbon monoxide from impinging flames. The preliminary findings of this more complete study shows that electric fields can reduce CO release from impinging flames when the flame is at the cusp of its separation from a closed to open tip flame, which, coincidentally, is the condition where maximum heat transfer from the flame is achieved.

Bibliography Type:	Description: (Last Updated: 11/26/2025)
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